

LUA — Implementation Status vs. Plan

Cross-check against "Lily Upskilling Academy.pdf" (the product plan)

Generated 2026-09-17 from a direct review of the current codebase (routes, server functions, database schema, and live database contents).
Structured to follow the plan document's own section order.

● Implemented ● Partial ● Not implemented ● Blocked (external dependency)

Overall: The core candidate journey (assessments → AI verification → certificates → badges) and almost the entire admin portal are implemented and live against a real Supabase database — well beyond a prototype. The two structural gaps are (1) **Bustler SSO** — fully spec'd but blocked on Bustler providing OAuth credentials, and (2) **fine-grained employee roles** — the plan's "verification officer / support executive / content manager" split doesn't exist; today it's just `admin` vs `reviewer`. A handful of admin features are real but noticeably simpler than the plan describes (certificate AI-confidence tiering, fraud detection) — flagged individually below.

1. Bustler Integration

Plan item	Status	Notes
Deep-link with pre-filled skill/category	Done	<code>/assessments?skill=&category=</code> — pre-fills, scrolls to and highlights the matching skill, shows a "Continuing your verification" banner.
Bustler account login (no separate registration)	Blocked	Fully specified in <code>docs/bustler-sso-integration.md</code> (signed-JWT handoff design) but not built — LUA has no OAuth client ID/signing key from Bustler yet. Candidates still sign up/in independently (email or Google).
Sync profile from Bustler	Blocked	Same blocker as above.
Reverse sync: badge events → Bustler webhook	Blocked	Call sites identified (<code>issueBadge</code> / <code>revokeBadgeAdmin</code>) but no webhook exists — needs Bustler's endpoint + public key.

2. Candidate Dashboard

Module	Status	Notes
Overview	Done	"Next action" guidance, quick stat tiles, recent certificates.
Skills	Done	Live inventory + (as of this review) an integrated "needs attention" verification summary.
Verification Progress	Done	Folded into the Skills page rather than a separate page (a deliberate simplification made during this review — see note below).
Assessment History	Done	Full attempt log with scores, pass/fail, resume-in-progress.
Certificates	Done	AI-issued + manual upload flow, status tracking.
Analytics (candidate-facing)	Done	<code>/analytics</code> exists for candidates in addition to the admin one.
Profile	Done	

Notifications	Done	Full feed page + bell dropdown; select-and-delete (dismiss) added in this review.
AI recommendation (Future)	Partial	Marked "(Future)" in the plan itself. A "Career Insights" widget exists but shows generic, hardcoded suggestions, not personalized AI output.

Note: earlier in this review, "My Skills," "Verification," and "Verification History" existed as three separate destinations with heavily overlapping content (same per-skill status, same action buttons). Verification Center was merged into My Skills to remove that redundancy; Verification History (the assessment-attempt log) kept its own page since it's a genuinely different thing.

3. Skill Management

Plan item	Status	Notes
Add / Remove / View skills	Done	
Request a skill not in the catalog	Done	Goes to an admin review queue (<code>skill_requests</code> table), approve/reject with notes.

4. AI Skill Verification

Plan item	Status	Notes
Beginner: 5 questions	Done	Matches exactly (<code>LEVEL_CONFIG</code>).
Expert: 15 questions	Done	Matches exactly.
AI Skill Verified badge on pass	Done	Issued only on an expert -level pass. A bug found and fixed in this review: a beginner-only pass was incorrectly being recorded as if it granted this badge, both in the candidate-facing assessments catalog and in admin analytics — now corrected.

5. Certificate Verification

Plan item	Status	Notes
Upload PDF/image as proof	Done	
Manual review by employees	Done	Approve/Reject with notes; "Request Re-upload" specifically is not a distinct action (candidate can just re-upload after a rejection).
Certification Verified badge on approval	Done	
AI confidence score + auto-verify high-confidence / duplicate check / tampering analysis	Not implemented	The review queue is manual-only for every submission — no AI confidence scoring, no auto-approval tier, no duplicate-detection or tampering analysis at upload time. This is the single largest gap between the plan and the built certificate flow.

6. Badge System

Plan item	Status	Notes
Per-skill badges (e.g. Electrician: AI + Certificate)	Done	Badges are scoped <code>(user, skill, type)</code> .
Fully Verified Bustler badge	Done	Computed (not stored) — true when every skill on the profile has ≥ 1 active verification. Shown on Skills page and Dashboard.
Institution Verified badge (Phase 2)	Placeholder only	Type wired through the badge system and UI (label, icon, admin filter) so it can be turned on later without a redesign, per the plan's own instruction — but no institution-partner workflow exists yet, correctly, since that's explicitly Phase 2.

7. Customer / Employer View

Plan item	Status	Notes
Public "View Verification" per skill/certificate	Done	<code>/verify/:id</code> — queries the real, live <code>certificates</code> table (score, verifier, issued/expiry, skills covered, unique certificate ID). Confirmed during this review to be live data, not the stale mock fixtures earlier documentation warned about.

8. Phase 2 (Future) — Institution Verification

Plan item	Status	Notes
Institution partner assessment + badge workflow	Correctly deferred	Plan explicitly scopes this to "after Lily Upskilling Academy partners with institutions" and asks only for architectural placeholders now — which exist (see Badge System above).

9. Admin Portal

9.1 User Management

Plan item	Status	Notes
Search/view any candidate profile	Done	
View Profile / History / Certificates	Done	
Suspend	Done	
Reset Assessment	Done	Resets an in-progress attempt.
Remove Badge	Done	
Login History	Not implemented	Not surfaced anywhere in the admin candidate-detail view.

9.2 Skill Management (admin)

Plan item	Status	Notes
Create / Edit / Delete skills	Done	
Review candidate-submitted new-skill requests	Done	

9.3 AI Assessment Management

Plan item	Status	Notes
View assessment details (skill, level, questions, score, duration, pass/fail)	Done	
Performance analytics (pass rate, avg score, attempts per skill)	Done	On the Analytics page.
Approve retakes for valid reasons	Done	Retake-request workflow with approve/reject.
Reset attempts for technical issues	Done	
Review suspicious attempts / flag	Done	Feeds the Fraud Signals page's "manually flagged" section.

9.4 Certificate Verification Center

Plan item	Status	Notes
Pending queue with user/skill/organization	Done	
Approve / Reject / Add comments	Done	"Request Re-upload" is not a distinct action.
AI confidence score, auto-verify at high confidence, duplicate check, tampering analysis	Not implemented	Same gap noted in section 5 — every certificate goes through the identical manual pending → approve/reject path regardless of any confidence signal, because no confidence scoring exists.

9.5 Badge Management

Plan item	Status	Notes
View/search all issued badges	Done	
Revoke	Done	
Reissue after re-verification	Done	
Handle badge appeals	Done	Full appeal submission + admin approve/reject flow.

9.6 Analytics Dashboard

Plan item	Status	Notes
Users monthly growth	Done	6-month chart.
Assessments pass rate	Done	
Popular skills (Top 10)	Done	
Average scores per category	Done	
Certificates: verified/rejected/pending	Done	
Badge distribution (incl. Fully Verified)	Done	

9.7 Fraud Detection Dashboard

Plan item	Status	Notes
Duplicate certificates	Done	
Repeated failed attempts	Done	
Manually flagged attempts	Done	
Suspicious uploads (behavioral)	Not implemented	
Multiple accounts	Not implemented	The page's own in-app disclaimer is explicit: "This isn't a fraud-detection model — there's no document forensics, device fingerprinting or duplicate-account detection here." Only the three signals above are computed from existing records; everything else the plan lists is genuinely absent.
Fake certificates (forensics)	Not implemented	
Edited documents (tampering detection)	Not implemented	

9.8 Support Center

Plan item	Status	Notes
Ticket list with user/issue/status/priority	Done	
Reply	Done	
Close	Done	Via status change to "closed".
Escalate	Done	Dedicated "escalated" status.

9.9 Reports

Plan item	Status	Notes
Daily / Weekly / Monthly reports	Done	"Activity report" with period selector.
Certificate report	Done	
Verification report	Done	Badge issuance/revocation export.

User Activity report	Done	
Fraud report	Done	Exports the same signals as the Fraud Dashboard (so inherits its scope limits above).
Format	Partial	All exports are CSV; the plan doesn't specify format so this is a reasonable interpretation, not a gap, but worth confirming if a specific format (e.g. PDF) was expected.

9.10 Notifications (admin)

Plan item	Status	Notes
"20 new certificates awaiting review"	Done	Equivalent pending-certificate alerts exist.
"New skill request submitted"	Done	
"User appealed badge rejection"	Done	
"AI confidence below 70%"	Not implemented	Depends on the AI-confidence scoring that doesn't exist (see 9.4).
"Potential duplicate certificate detected"	Partial	Detectable on the Fraud page, but not pushed as a proactive notification.
Dedicated notifications page + sidebar entry, select & delete	Done	Added during this review — previously bell-dropdown only.

9.11 Audit Logs & Employee Roles

Plan item	Status	Notes
Audit log (employee, action, affected user, timestamp)	Done	Admin → Settings.
Role-based access (verification officers, support executives, content managers each scoped to their own modules)	Not implemented	Current roles are just <code>admin</code> (full access) and <code>reviewer</code> — no scoping of which admin nav sections a given employee can see/use by job function. This is the second major structural gap alongside Bustler SSO.

10. Known platform-level issue (not a plan gap)

This project's dev/build toolchain runs on beta releases (`vite@8.1.4` , `nitro@3.0.260603-beta` , pre-1.0 `@tanstack/react-start`). Several distinct instability bugs were traced back to this combination during this review (a client-side `process.env` define failure, dependency-optimizer reload glitches, an SSR/hydration race). Worth flagging separately from feature completeness since it affects reliability, not scope.